

Banking Dashboard - End-to-End Data Analysis Project

Project Overview

This project analyzes banking customer data using Python and Power BI to deliver business insights through data cleaning, analysis, and visualization.

Business Problem

Analyze customer deposits, loans, income, and demographics to support better banking decisions.

Tools & Technologies

Python (Pandas, NumPy, Matplotlib, Seaborn), Power BI, Power Query, DAX, CSV Dataset.

Project Workflow

CSV Dataset → Data Cleaning → Exploratory Data Analysis → Power BI Dashboard → Business Insights

Dashboard Pages

- Home
- Loan Analysis
- Deposit Analysis
- Summary

Recommended KPIs

- Total Customers
- Total Deposits
- Total Loans
- Average Income
- Average Credit Card Balance
- Total Savings
- Total Checking
- Total Business Lending

Key Business Insights

- Higher-income customers tend to maintain larger deposits.
- Deposit-related accounts are positively correlated.
- Business lending customers generally maintain higher balances.
- Customer demographics influence product usage.

Repository Structure

```
Banking-Analytics/  
├── BankEDA.ipynb  
├── Banking.csv  
├── Banking_Dashboard.pbix  
├── README.md  
├── requirements.txt  
├── LICENSE  
└── images/
```

Dashboard Screenshots

The README references dashboard images. Ensure the following files exist in the repository:

```
images/  
  home.png  
  loan_analysis.png  
  deposit_analysis.png  
  summary.png
```

Then update the README image paths as:

```
  
  
  

```

Future Improvements

SQL Integration, Machine Learning, Customer Churn Prediction, Loan Default Prediction, Azure Deployment.

Conclusion

This project demonstrates a complete end-to-end data analytics workflow suitable for a Data Analyst portfolio.